Inferring Spatial Connectivity Decay Exponent from Graph Topology Alone

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Abstract

A spatially embedded neuronal network is constructed by sampling connections with probability proportional to $D_{ij}^{-\delta_{\rm ker}}$, where D_{ij} is the Euclidean distance between neurons i and j and $\delta_{\rm ker} \in [0, 3]$ is a real-valued decay exponent. The limiting cases interpolate between uniform random connectivity ($\delta_{\rm ker} = 0$) and strongly local, lattice-like wiring ($\delta_{\rm ker} \to 3$). We ask: given only the binary adjacency matrix G — without access to neuron positions or distances — can one estimate $\delta_{\rm ker}$? We present five purely graph-theoretic diagnostics, each producing a single scalar that varies monotonically with $\delta_{\rm ker}$ and therefore serves as a proxy for the hidden spatial decay. The common principle is that large $\delta_{\rm ker}$ yields a lattice-like graph (high local clustering, assortative degrees, low spectral dimension), while small $\delta_{\rm ker}$ yields a small-world or Erdős–Rényi-like graph (diffuse neighborhoods, weaker clustering, higher effective dimensionality).

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1 Setup and Notation

1.1 The Generative Model

Consider N neurons placed on a regular two-dimensional grid within a rectangular domain $[0, L] \times [0, H]$ with minimum spacing $d_{\min} > 0$. After random subsampling of grid sites and lexicographic sorting, each neuron j occupies a position $\mathbf{P}_j = (x_j, y_j)$. The pairwise Euclidean distance matrix is $D_{ij} = \|\mathbf{P}_i - \mathbf{P}_j\|_2$.

Connections are drawn according to a distance-dependent affinity kernel. For every ordered pair (i, j) with $i \neq j$, define the affinity

$$V_{ij} = D_{ij}^{-\delta_{\mathrm{ker}}}, \qquad i \neq j; \qquad V_{ii} = 0, \tag{1}$$

where $\delta_{\mathrm{ker}} \in [0, 3]$ is a real-valued *decay exponent* that controls how sharply connection probability falls off with distance. Because grid nodes are separated by at least $d_{\min} > 0$, the affinity is everywhere finite.

The exponent δ_{ker} interpolates continuously between qualitatively distinct connectivity regimes:

δ_{ker}	Character of the resulting graph
0	Erdős–Rényi: connection probability independent of distance
$\lesssim 1$	Weak spatial bias; long-range edges abundant; small-world-like
≈ 1	Moderate locality; intermediate clustering
≈ 2	Strong locality; neighborhood structure clearly visible
$\gtrsim 2$	Very tight spatial coupling; lattice-like communities
3	Extreme locality; connections almost exclusively between nearest neighbors

Each presynaptic neuron j independently samples an integer out-degree $k_j \sim \mathrm{Uniform}(k_{\min}, k_{\max})$ and draws k_j distinct postsynaptic targets $\mathcal{T}_j \subset \{1, \ldots, N\} \setminus \{j\}$ without replacement from the normalized distribution

$$p_{ij} = \frac{V_{ij}}{\displaystyle\sum_{i' \neq j} V_{i'j}} = \frac{D_{ij}^{-\delta_{\mathrm{ker}}}}{\displaystyle\sum_{i' \neq j} D_{i'j}^{-\delta_{\mathrm{ker}}}}, \qquad i \neq j. \tag{2}$$

1.2 The Inference Problem

We are given only the weighted connectivity matrix $A_{\mathrm{true}} \in \mathbb{R}^{N \times N}$. Since A-matrix represents interaction between neurons which can be either excitatory or inhibitory, the values of A_{ij} can be either positive or negative.

Neuron positions $\mathbf{P}$, pairwise distances D, and the value of δ_{ker} are all *unknown*. From A_{true} we extract the binary adjacency matrix

$$G_{ij} = \begin{cases} 1 & A^{\mathrm{off}}_{\mathrm{true},ij} \neq 0,\ i \neq j, \\ 0 & \text{otherwise}, \end{cases} \tag{3}$$

where A^{off} denotes the off-diagonal part of A_{true} (self-loops are excluded). The goal is to estimate the decay exponent δ_{ker} from G alone, or at minimum to construct scalar diagnostics that vary monotonically with δ_{ker} and therefore distinguish, e.g., $\delta_{\mathrm{ker}} = 1$ from $\delta_{\mathrm{ker}} = 2$.

The key qualitative insight is that as δ_{ker} increases, the graph becomes more spatially clustered: edges are shorter, neighborhoods overlap more, the graph looks more like a lattice. Conversely, as δ_{ker} decreases toward zero, edges span the full domain, neighborhoods are diffuse, and the graph approaches an Erdős–Rényi random graph. Each of the five methods below detects a different combinatorial or spectral consequence of this transition.

1.3 Additional Notation

Throughout, we use the following derived quantities from G:

$$k_i^{\text{in}} = \sum_j G_{ij}, \qquad k_j^{\text{out}} = \sum_i G_{ij}, \qquad |\mathcal{E}| = \sum_{i,j} G_{ij}, \tag{4}$$

denoting the in-degree and out-degree of each neuron and the total number of directed edges. The symmetrized adjacency is $\tilde{G} = (G + G^\top)/2$ with degree matrix $\tilde{D}_{ii} = \sum_j \tilde{G}_{ij}$. Neighborhoods are defined as $\mathcal{N}_{\text{in}}(i) = \{j : G_{ij} = 1\}$ and $\mathcal{N}_{\text{out}}(j) = \{i : G_{ij} = 1\}$.

2 Method 1: Degree Assortativity

2.1 Definition

The Newman assortativity coefficient measures the Pearson correlation between the degrees at either end of an edge. For the directed graph G, define the in-degree of the target and the in-degree of the source for each edge $(i, j) \in \mathcal{E}$ (meaning $G_{ij} = 1$, i.e., neuron j drives neuron i):

$$r = \frac{|\mathcal{E}|^{-1} \displaystyle\sum_{(i,j)\in\mathcal{E}} k_i^{\text{in}}\, k_j^{\text{in}} \;-\; \left[|\mathcal{E}|^{-1} \displaystyle\sum_{(i,j)\in\mathcal{E}} k_i^{\text{in}}\right] \left[|\mathcal{E}|^{-1} \displaystyle\sum_{(i,j)\in\mathcal{E}} k_j^{\text{in}}\right]}{\sigma_{\text{in,target}}\;\sigma_{\text{in,source}}}, \tag{5}$$

where $\sigma_{\text{in,target}}$ and $\sigma_{\text{in,source}}$ are the standard deviations of $\{k_i^{\text{in}}\}_{(i,j)\in\mathcal{E}}$ and $\{k_j^{\text{in}}\}_{(i,j)\in\mathcal{E}}$ over all edges.

2.2 Dependence on δ_{ker}

When δ_{ker} is large, each neuron's targets are concentrated in a tight spatial neighborhood. A neuron with high in-degree sits in a locally dense region and receives input predominantly from nearby neurons, which themselves receive input from the same dense region — hence they too have high in-degree. High-degree nodes connect to high-degree nodes, producing *assortative* mixing ($r > 0$).

As δ_{ker} decreases, edges reach farther, a neuron's targets span diverse spatial neighborhoods with diverse degree profiles, and the degree–degree correlation weakens. In the limit $\delta_{\text{ker}} \to 0$ (Erdős–Rényi), degrees are independent of position and $r \to 0$.

The assortativity coefficient thus increases monotonically with δ_{ker}:

$$\delta_{\text{ker}}^{(1)} < \delta_{\text{ker}}^{(2)} \quad \Longrightarrow \quad r^{(1)} \lesssim r^{(2)}. \tag{6}$$

3 Method 2: Common-Neighbor Overlap (Jaccard Index)

3.1 Definition

For each directed edge $j \to i$ (i.e. $G_{ij} = 1$), the Jaccard overlap between the in-neighborhood of the target and the out-neighborhood of the source is

$$J_{ij} = \frac{|\mathcal{N}_{\text{in}}(i) \cap \mathcal{N}_{\text{out}}(j)|}{|\mathcal{N}_{\text{in}}(i) \cup \mathcal{N}_{\text{out}}(j)|}. \tag{7}$$

This answers a purely combinatorial question: *if j drives i, how many of j's other targets also drive i?* The mean Jaccard index over all edges is

$$\bar{J} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} J_{ij}. \tag{8}$$

3.2 Dependence on δ_{ker}

The Jaccard index is the graph-theoretic analog of spatial locality without ever defining locality explicitly. When δ_{ker} is large, each neuron's fan-out lands in a compact spatial neighborhood; two neurons connected by an edge are necessarily close, so their neighborhoods overlap substantially and J_{ij} is large. When δ_{ker} is small, fan-out is diffuse, common neighbors between any linked pair become rare, and $\bar{J}$ shrinks. In the Erdős–Rényi limit ($\delta_{\text{ker}} = 0$) with edge density p, the expected Jaccard index scales as $\bar{J} \sim p/(2-p)$, independent of spatial structure.

The mean Jaccard index therefore increases monotonically with δ_{ker}:

$$\delta_{\text{ker}}^{(1)} < \delta_{\text{ker}}^{(2)} \quad \Longrightarrow \quad \bar{J}^{(1)} \lesssim \bar{J}^{(2)}. \tag{9}$$

4 Method 3: Cycle Closure Decay Rate

4.1 Definition

Powers of the adjacency matrix count directed walks: $(G^k)_{ij}$ is the number of distinct walks of length k from j to i. In particular, $\text{tr}(G^{k+1})$ counts closed walks (directed cycles) of length $k+1$, while $\mathbf{1}^\top G^k \mathbf{1} = \sum_{i,j}(G^k)_{ij}$ counts all walks of length k. Define the *k-step transitivity* as the fraction of length-k walks whose endpoints are connected by a direct edge (thus closing into a $(k+1)$-cycle):

$$\mathcal{T}_k = \frac{\text{tr}(G^{k+1})}{\mathbf{1}^\top G^k \, \mathbf{1}}, \qquad k = 1, 2, \ldots \tag{10}$$

At $k = 1$ this reduces (up to normalization) to the directed clustering coefficient. The *cycle closure decay rate* is the log-slope of $\mathcal{T}_k$ over a short range:

$$\xi = -\frac{d \log \mathcal{T}_k}{dk}\bigg|_{k=2,\ldots,K}, \tag{11}$$

estimated by linear regression of $\log \mathcal{T}_k$ on k for $k = 2, \ldots, K$ (typically $K = 5$ or 6).

4.2 Dependence on δ_{ker}

For a graph embedded in two dimensions, the existence of short cycles is a signature of spatial locality: a walk that stays in a compact region is far more likely to return to its origin than one that wanders across the domain. When δ_{ker} is large, neighborhoods are tight and densely interconnected, so walks of moderate length close into cycles readily — $\mathcal{T}_k$ remains high across k and ξ is small.

When δ_{ker} is small, long-range edges carry walks far from their starting point. A two-step walk $j \to i \to i'$ may traverse the full domain, and the probability that i' connects back to j is no higher than the background edge density. Cycle closure becomes increasingly unlikely with each additional step, so $\mathcal{T}_k$ drops steeply and ξ is large.

The cycle closure decay rate therefore increases monotonically as δ_{ker} decreases:

$$\delta_{\mathrm{ker}}^{(1)} < \delta_{\mathrm{ker}}^{(2)} \quad \Longrightarrow \quad \xi^{(1)} \gtrsim \xi^{(2)}. \tag{12}$$

5 Method 4: Persistent Homology

This metric measures the summed lifetime of all loop features in the graph while sweeping from strong to weak edge weights.

We probe the topological complexity of the graph by computing its persistent homology. The symmetrized adjacency matrix $\tilde{G}_{ij} = \frac{1}{2}(G_{ij} + G_{ji})$ removes any directionality inherited from the original interaction matrix. From $\tilde{G}$ we define edge weights $w_{ij} = |\tilde{G}_{ij}|$ (interaction strength, irrespective of sign) and construct a filtered Vietoris–Rips complex $\mathcal{R}(\epsilon)$: a simplex $[v_0, \ldots, v_k]$ is included whenever every pairwise weight exceeds the threshold,

$$[v_0, \ldots, v_k] \in \mathcal{R}(\epsilon) \quad \Longleftrightarrow \quad w_{v_i v_j} \geq \epsilon \;\; \forall\, 0 \leq i < j \leq k. \tag{13}$$

Sweeping ϵ from $\max_{ij} w_{ij}$ down to zero produces a nested filtration; the persistence diagram records the birth–death pairs (b_i, d_i) for each homological feature that appears and subsequently merges or closes.

Two scalar summaries are extracted. The *total persistence* in dimension k,

$$\Pi_k = \sum_i (d_i - b_i), \tag{14}$$

measures the cumulative lifetime of all k-dimensional features. The *persistent Betti curve* $\beta_k(\epsilon)$ counts the number of features alive at filtration value ϵ; its integral $B_k = \int \beta_k(\epsilon)\, d\epsilon$ gives a single-number summary of topological richness.

For graphs embedded in two dimensions, only $k = 1$ (loops) carries meaningful geometric information; higher-dimensional features such as enclosed cavities ($k = 2$) are not expected in a planar embedding, and their computation requires enumerating $(k{+}1)$-cliques, which becomes prohibitively expensive for large graphs. We therefore restrict attention to Π_1.

For lattice-like graphs ($\delta_{\mathrm{ker}} \geq 2$), the dimension-1 barcode shows persistent loops reflecting the regular mesh, giving moderate Π_1. In the Erdős–Rényi limit ($\delta_{\mathrm{ker}} = 0$), random long-range edges create many short-lived loops, producing a cloud of short bars in Π_1 with no dominant persistent features.

Π_1 thus offers a purely topological diagnostic of the lattice-to-random transition, complementing the combinatorial and spectral methods of Sections 2–3.

6 Summary and Comparison

The four diagnostics are collected in Table 1. Each uses only the adjacency matrix G (or equivalently the zero pattern of $A_{\text{true}}^{\text{off}}$) unless otherwise noted. None requires neuron positions, pairwise distances, or spike data.

§	Method	Scalar	$\delta_{\text{ker}}\uparrow$	Weights	Computational cost
2	Degree assortativity	r	$\uparrow$	no	$O(\lvert\mathcal{E}\rvert)$
3	Jaccard overlap	$\bar{J}$	$\uparrow$	no	$O(\lvert\mathcal{E}\rvert\, k_{\max})$
4	Cycle closure decay	ξ	$\downarrow$	no	$O(K\,N^3)$ or eigendecomp
5	Persistent homology	Π_1	$\Pi_1\uparrow$	optional	$O(n_{\text{simp}}^3)$ reduction

Table 1: Summary of the four topological diagnostics for the decay exponent δ_{ker}. The arrow in the fifth column indicates the direction of change as δ_{ker} increases (graph becomes more spatially local). The "Weights" column indicates whether the method uses the magnitudes $|A_{\text{true},ij}|$ beyond the binary connectivity pattern. For persistent homology, n_{simp} denotes the number of simplices in the filtered complex.

Methods 1–3 operate exclusively on the binary adjacency matrix G and are entirely insensitive to the magnitudes of the interaction weights $A_{\text{true},ij}$. Degree assortativity and Jaccard overlap count edges and shared neighbors; cycle closure decay works with powers of G, both of which depend only on which entries are nonzero.

Persistent homology (Method 4) is the only diagnostic that can optionally exploit weight magnitudes. As formulated in Section 5, the filtration is built from $w_{ij} = |\tilde{G}_{ij}|$, where $\tilde{G}$ is the binary symmetrized adjacency. Because $\tilde{G}_{ij} \in \{0, \frac{1}{2}, 1\}$, this filtration has only two nontrivial thresholds and is effectively binary. A richer filtration is obtained by replacing the edge weights with the actual interaction strengths, $w_{ij} = \frac{1}{2}(|A_{\text{true},ij}|+|A_{\text{true},ji}|)$, so that strongly coupled pairs enter the complex before weakly coupled ones. This weighted variant introduces a continuous filtration parameter and may sharpen the topological signatures, at the cost of depending on weight magnitudes that the other three methods ignore.

The four methods probe complementary facets of the same underlying phenomenon. Degree assortativity (Method 1) detects correlations at the scale of single edges. Jaccard overlap (Method 2) examines two-hop neighborhood structure. Cycle closure decay (Method 3) interpolates between local and global scales through the parameter k. Persistent homology (Method 4) captures the topological skeleton of the graph — its loop structure — as a function of interaction strength, providing a diagnostic that is invariant under graph isomorphism and insensitive to node labeling.

For practical estimation of $\delta_{\text{ker}} \in [0, 3]$, one may generate a calibration family of synthetic graphs at a grid of δ_{ker} values (with the same N, $k_{\min}$, $k_{\max}$, and E/I ratio as the target network), compute all four scalars for each, and fit monotone calibration curves. Given an observed graph, the four diagnostics are evaluated and mapped through the calibration curves to produce four independent estimates of δ_{ker}, which can be combined (e.g., by averaging or by inverse-variance weighting) into a single consensus estimate.